MACHINE LEARNING (ML) / AI IN PUBLIC SECTOR / CRITICAL DATA STUDIES (CDS)

# PREDICTING HIGH
# CRIME AREAS IN T

## A synthetic data proof-of-concept for ML-based decision

## PROBLEM

Police resource allocation is often reactive, while real crime data is restricted by privacy, ethics, and access rules.

**Objective:** test whether a simple ML pipeline can predict high-risk area-time combinations without using real personal data.

**Core question:** can predictive policing systems be built technically, and what risks emerge when they guide decisions?

## METHOD

Unit of analysis: one Tartu district at one hour. Output: probability of the area-hour being high-risk.

Synthetic data > Feature engineering > ML comparison > Model choosing > Application

## DATA

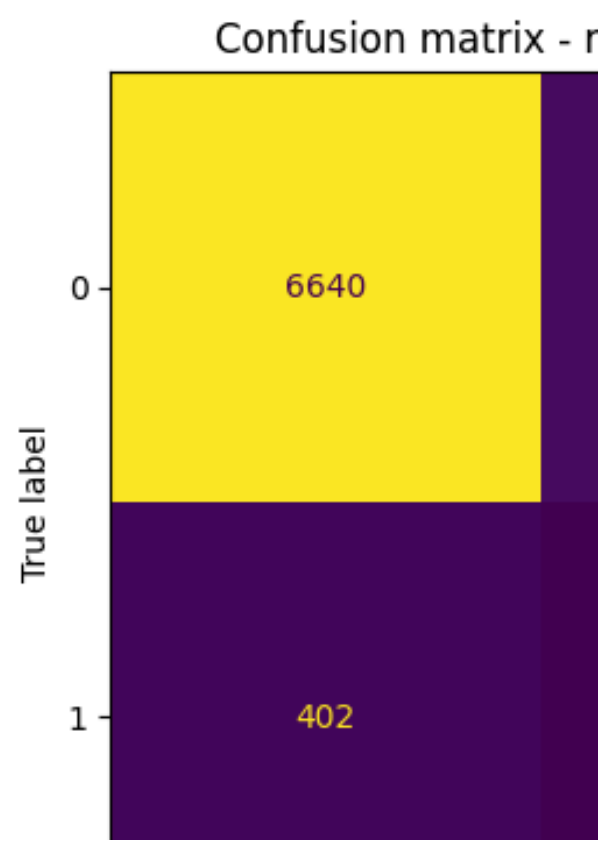
| | |
|---|---|
| **Environment** | 18 districts x 90 days x 24 hours = 38,880 area-hours. |
| **Target** | High-risk = incident_count >= 1; 3,297 rows (8.48%) are high-risk. |
| **Time** | hour, weekday, weekend, night |

## RESULTS

Random forest was the s
classifier by F1. Accuracy
area-hours are normal; p
reliable a risk flag is, reca
high-risk hours are caugh

| | |
|---|---|
| **88.7%** Random forest accuracy | **0.369** Random fo |
| **0.350** Random forest precision | **0.390** Random fo recall |

## CONFUSION M



# H-RISK
# TARTU

**support**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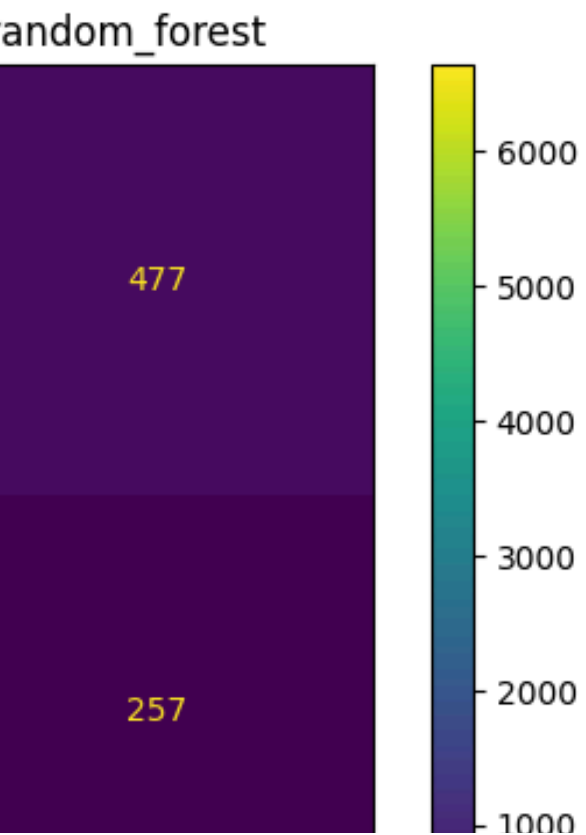
trongest high-risk
is high because most
recision shows how
ll shows how many
nt, and F1 balances both.

| | |
|---|---|
| **9**<br>orest F1 | **0.812**<br>Best ROC-AUC, logistic |
| **0**<br>orest | **38,880**<br>Area-hour rows |

## ATRIX RANDOM FOREST




## LIMITATIONS

- Synthetic data simplifies real-world complexity.
- The model predicts patterns, not causes.
- Inequality and segregation are missing or simplified.
- Thresholds turn uncertainty into concrete action.

## APPLICATION

- Interactive Tartu risk map with district-level scores.
- Selected suburb view shows 48-hour predicted incidents and risk.
- Prototype links model details, dataset view, policing plan, and critical analysis.

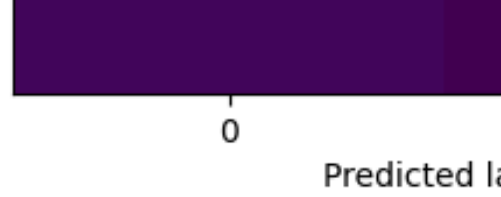
| | |
|---|---|
| **Context** | weather and city events |
| **Area** | density, nightlife, student zones |
| **History** | past incident patterns |

| | |
|---|---|
| **Why synthetic?** | No privacy risk, full control, reproducible experimentation. |
| **Weak point** | The model learns the assumptions used to create its reality. |

# MODELS TESTED

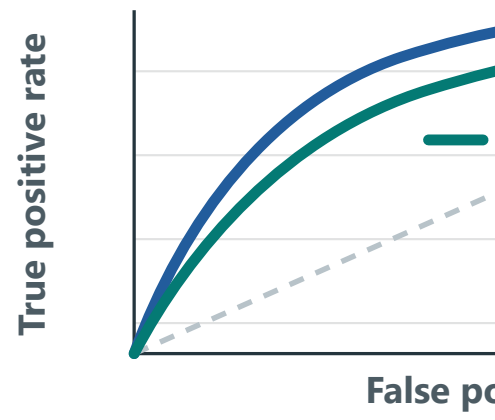
| Model | Task | Main setup | Notebook result |
|---|---|---|---|
| **Linear regression** | Count prediction | One-hot encoded features; 80/20 train-test split. | MAE 0.164, MSE 0.105, R2 0.146. Weak fit for incident counts. |
| **PyCaret regression sweep** | Count prediction | 19 regressors, 10-fold CV, session 845; best was gradient boosting. | Best R2 0.197, RMSE 0.316, MAE 0.149. Counts remained hard to explain. |
| **Decision tree** | Incident-count class | Default DecisionTreeClassifier, no depth limit. | Accuracy 0.894, macro-F1 0.225. Mostly learned the zero-incident majority. |
| **Logistic regression** | High-risk binary | Balanced class weights, max_iter 1000, probability scores. | Acc 0.749, precision 0.217, recall 0.753, F1 0.337, ROC-AUC 0.812. |
| **Random forest** | High-risk binary | 100 trees, min_samples_leaf 2, balanced_subsample, thresholded labels. | Acc 0.887, precision 0.350, recall 0.390, F1 0.369, ROC-AUC 0.768. Best F1. |

**Takeaway:** classification was more useful than count regression; random forest gave the best F1, while logistic regression ranked risk best by ROC-AUC.



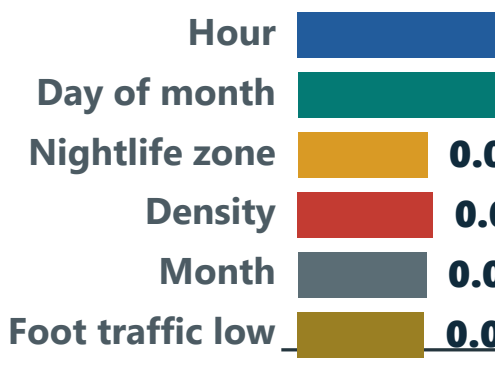

**How to read:** Rows show the re
model prediction. Label 0 means
risk. Correct predictions are 6,64
high-risk area-hours. Errors are 4
high-risk cases.

# ROC CURVE




**ROC reading:** The ROC curve sh
catching high-risk area-hours an
Logistic regression ranked cases
random forest was lower on ROC
best on F1, so it is used for the c

# FEATURE IMPORTANCE



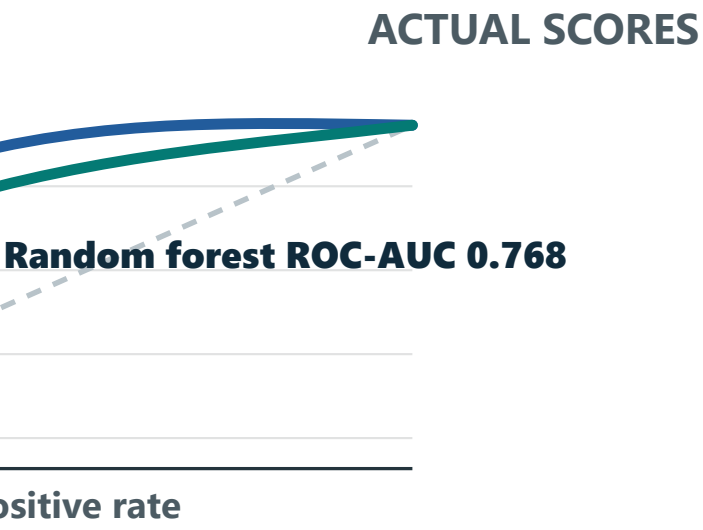

**Feature insight:** Top features ar
strong signals, strong assumptio

## References

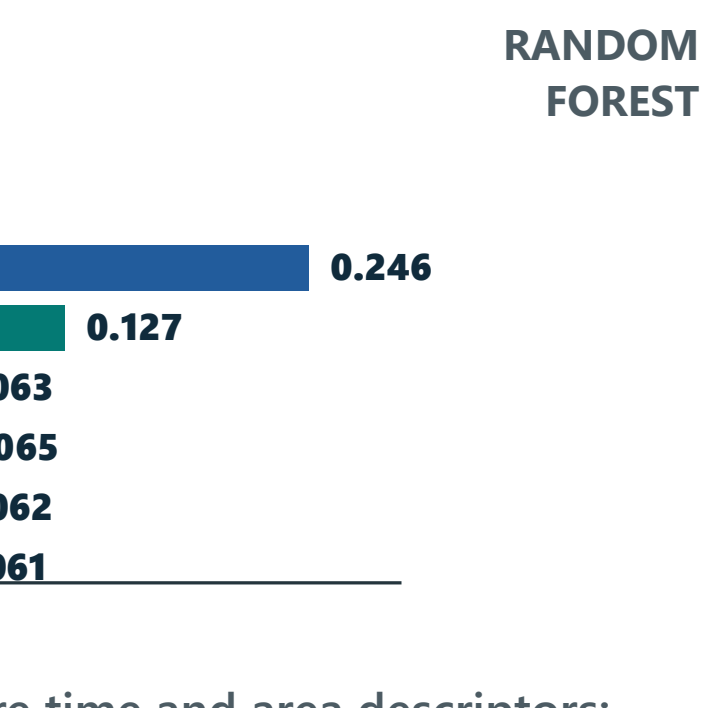
**Project materials in GitHub:** Projekt (1).ipynb; tartu_risk_m

**Critical frame:** Meijer, A., & Wessels, M. (2019). Predictive Poli
of Public Administration, 42(12), 1031-1039. https://doi.org/10

al label; columns show the
s normal, label 1 means high-
0 normal area-hours and 257
477 false alarms and 402 missed




ows the trade-off between
d falsely flagging normal ones.
best overall (ROC-AUC 0.812);
C-AUC (0.768) but performed
onfusion matrix above.




re time and area descriptors:
ns.

# CRITICAL ANALYSIS

**Operational logic:** identify top-risk district-hours and allocate more patrol attention there.

**More patrols** > **More detected incidents**

**Higher risk score** > **More patrols again**

**Critical point:** the system does not detect crime; it detects where police look.

- **Datafication:** crime becomes a pattern, not a social phenomenon.
- **Feedback loops:** repeated attention can manufacture hotspots.
- **Bias amplification:** data structures can reproduce inequality.
- **False objectivity:** scientific-looking outputs hide assumptions.

# CONCLUSION

**Best model:** random forest was the most useful classifier for high-risk area-hours.

**Main limitation:** incident-count prediction was weak because high-risk rows are rare.

**ML takeaway: engineered time, area, context, and history features can rank synthetic risk better than chance, but the prototype needs stronger validation before any real-world use.**

l_models-1.ipynb; A1 poster
cing: Review of Benefits and Drawbacks. International Journal
.1080/01900692.2019.1575664